

Project Report

Library Management System

Developed using Python

1. Introduction

This report documents the design, implementation, and working of a Library Management System developed using Python. The project is a console-based application that allows management of a library's book collection through simple menu-driven operations.

2. Objective

The objective of this project is to build a functional, easy-to-use system that automates basic library tasks such as adding, viewing, searching, issuing, returning, deleting, and counting books, replacing manual record-keeping with a simple software solution.

3. Tools and Technologies Used

Component	Details
Programming Language	Python 3
Data Structure	List (books = [])
Interface	Command-Line Interface (CLI)
Control Flow	while loop, if-elif-else conditional statements

4. System Design

4.1 Data Storage

The system uses a single Python list, **books**, to store the names of all available books. Each book is stored as a string element in the list.

4.2 Main Program Flow

The program runs inside an infinite while-True loop that:

- Displays a menu of 8 options.
- Accepts the user's choice as input.
- Executes the corresponding block of code based on the choice.
- Repeats until the user selects the "Exit" option, which breaks the loop.

4.3 Module-wise Description

Option	Function	Logic
1. Add Book	Adds a new book	Takes book name as input and appends it to the books list.
2. View Books	Displays all books	Checks if the list is empty; if not, iterates and prints each book.
3. Search Book	Checks availability	Uses the 'in' operator to check if the book exists in the list.

4. Issue Book	Issues a book	If the book exists, removes it using <code>.remove()</code> , simulating it being taken out.
5. Return Book	Returns a book	Appends the book back into the list, making it available again.
6. Delete Book	Deletes a book	If present, removes the book permanently from the list.
7. Count Books	Counts total books	Uses <code>len()</code> to display the number of books currently in the list.
8. Exit	Ends the program	Breaks out of the while loop, ending execution.

5. Working / Methodology

The program follows a simple linear workflow:

- The user is shown a menu each time the loop runs.
- Based on the numeric input, the corresponding `elif` block executes.
- All operations directly manipulate the single books list — no separate database or file is used.
- Invalid menu choices are handled with an `else` clause that prompts the user to try again.

6. Sample Output

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return book
6. Delete book
7. Count books
8. Exit
Enter your choice: 1
Enter book name: Python Basics
book get added successfully
```

7. Testing

The following scenarios were considered during testing:

- Adding a book and verifying it appears under "View Books".
- Searching for a book that exists vs. one that does not.
- Issuing a book that is available, and attempting to issue one that is not.
- Returning a book after issuing it, and confirming it reappears in the list.
- Deleting a book and confirming it no longer appears in searches.
- Counting books after multiple add/delete/issue operations.
- Entering an invalid menu option to confirm the error message is shown.

8. Limitations

- Data is not saved permanently; it resets every time the program is closed since it is stored only in memory.
- No unique identifiers for books — two books with the same name are treated as identical entries.
- No borrower information (name, issue date, due date) is tracked when a book is issued.
- No input validation for empty or duplicate book names.

9. Future Scope

- Add file handling or a database (e.g., SQLite) for permanent data storage.
- Introduce borrower records with issue/return dates and fine calculation for late returns.
- Assign a unique ID to each book to avoid ambiguity with duplicate titles.
- Develop a Graphical User Interface (GUI) using Tkinter or a web-based interface.

10. Conclusion

This project successfully demonstrates the core concepts of Python programming — lists, loops, conditional statements, and functions — applied to a real-world use case. The Library Management System, while simple, fulfills the essential requirements of tracking and managing a library's book inventory through an interactive command-line interface.